Marginalizing shocks

Cover that shocks which I ought not to see.

Model reparameterization

We marginalize the shocks in the mixture model! Going from:

```
# Non-shocky
target += normal_lpdf(y[n] | mu[n] + delta_sck[n], sigma)
target += normal_lpdf(delta_sck[n] | 0, inner_tau_sck)

# Shocky
target += normal_lpdf(y[n] | mu[n] + delta_sck[n], sigma)
target += normal_lpdf(delta_sck[n] | 0, outer_tau_sck)
```

to a log-likelihood where shocks are marginalized out:

```
# Non-shocky
target += normal_lpdf(y[n] | mu[n], eta)

# Shocky
target += normal_lpdf( y[n] | mu[n], eta * sqrt{1 + nu / eta^2})
```

where:

$$\eta = \sqrt{\sigma^2 + {\tau_{\text{inner}}^{\text{shock}}}^2}$$

$$\nu = {\tau_{\text{outer}}^{\text{shock}}}^2 - {\tau_{\text{inner}}^{\text{shock}}}^2$$

In Stan, we now have in the model block:

```
writeLines(
  readLines(file.path(wd, 'model', 'stan',
                      'model9_marginalshocks.stan'))[230:315])
```

```
model {

  alpha ~ normal(0, 0.69); // 0.2 mm <~ exp(alpha) * 1 mm <~ 5 mm

  beta_gdd ~ normal(0, log(1.8) / 2.57); // -log(1.8) <~ beta_gsl <~ log(1.8)
  beta_sm ~ normal(0, log(1.8) / 2.57); // -log(1.8) <~ beta_gsl <~ log(1.8)
  beta_vpd ~ normal(0, log(1.8) / 2.57); // -log(1.8) <~ beta_gsl <~ log(1.8)

  rho_sp ~ lognormal(2.65, 0.135); // 10 <~ rho <~ 20

  gamma_sp ~ normal(0, log(10) / 2.57); // 0 <~ gamma <~ log(10)

  for (s in 1:N_stands)
    f_tilde_sh[s] ~ normal(0, 1);
  rho_sh ~ lognormal(1.7, 0.26);       // 3 <~ rho_sh <~ 10
  gamma_sh ~ normal(0, log(3) / 2.57); // 0 <~ gamma_sh <~ log(3)

  eta ~ gamma(6, 43.5);
```

```
  nu ~ gamma(0.5, 0.033);
  phi_sck ~ beta(2, 20);
  omega_conc_sck ~ beta(230, 14); // 0.9 <~ omega_conc_sck <~ 0.97 (most trees, but not ALL trees)
  omega_nonconc_sck ~ beta(1, 20); // 0 <~ omega_conc_sck <~ 0.15 (should be rare, but... who knows?)

  vector[N] mu;
  for (t in 1:N_trees) {
    array[N_years[t]] int tree_idxs
    = linspaced_int_array(N_years[t],
                          tree_start_idxs[t],
                          tree_end_idxs[t]);

    array[N_years[t]] int all_years_idxs_tree = all_years_idxs[tree_idxs];
    array[N_years[t]] real years_tree = years[tree_idxs];

    int stand_idx = stand_idxs[t];
    int species_idx = species_idxs[t];

    vector[N_years[t]] gdd_obs_tree = gdd_obs[tree_idxs];
    vector[N_years[t]] sm_obs_tree = sm_obs[tree_idxs];
    vector[N_years[t]] vpd_obs_tree = vpd_obs[tree_idxs];

    mu[tree_idxs] =  alpha
    + beta_gdd[species_idx] * (gdd_obs_tree - gdd0)
    + beta_sm[species_idx] * (sm_obs_tree - sm0)
    + beta_vpd[species_idx] * (vpd_obs_tree - vpd0)
    + f_sh[stand_idx, all_years_idxs_tree];

    f_tilde[tree_idxs] ~ normal(0,1);
  }

  // Observational model with marginalized shocks
  for (s in 1:N_stands) {

    array[N_stand_trees[s]] int stand_trees_idxs = linspaced_int_array(N_stand_trees[s],
      stand_trees_start_idxs[s], stand_trees_end_idxs[s]);
    vector[N_stand_years[s]] log_p0 = rep_vector(0, N_stand_years[s]);
    vector[N_stand_years[s]] log_p1 = rep_vector(0, N_stand_years[s]);

    for(ts in 1:N_stand_trees[s]){
      int t = stand_trees_idxs[ts];
      array[N_years[t]] int tree_idxs = linspaced_int_array(N_years[t], tree_start_idxs[t], tree_end_idxs[t]);
      array[N_years[t]] int all_years_idxs_tree = all_years_idxs[tree_idxs];

      for(y in 1:N_years[t]) {
        int ys = all_years_idxs_tree[y]-stand_start_years_idxs[s]+1;

        log_p0[ys] += log_mix(omega_nonconc_sck,
                          normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[tree_idxs[y]]
                          + f[tree_idxs[y]], eta * sqrt(1 + nu / eta^2)),
                          normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[tree_idxs[y]]
                          + f[tree_idxs[y]], eta));
        log_p1[ys] += log_mix(omega_conc_sck,
                          normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[tree_idxs[y]]
                          + f[tree_idxs[y]], eta * sqrt(1 + nu / eta^2)),
                          normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[tree_idxs[y]]
                          + f[tree_idxs[y]], eta));

      }
    }

    for(y in 1:N_stand_years[s]) {
      target += log_mix(phi_sck[s], log_p1[y], log_p0[y]);
    }
  }

}
```

But... my shocks...

Our large shocks are likely tied to interesting ecological stuff... But by marginalizing them, we don't have inferences on them anymore.
It could be doable to "*to marginalize out eta in the "small shock" branch of the mixture model but leave eta explicit in the "large shock" branch of the mixture model*" (Mike).
Since I'm not totally sure how to do that, I decided to sample the δ_{shock} afterwards, to recover their posterior distributions in the generated quantities block. This is an ongoing project, so for now I recover only the shock state (z_t).

```
writeLines(
  readLines(file.path(wd, 'model', 'stan',
                      'model9_marginalshocks.stan'))[351:405])
```

```
generated quantities {
  array[N] int sck_state; // latent state, zt = 0 or zt = 1
  array[N] real log_rw_pred;
  for (t in 1:N_trees) {
    array[N_years[t]] int tree_idxs
    = linspaced_int_array(N_years[t],
                          tree_start_idxs[t],
                          tree_end_idxs[t]);
    array[N_years[t]] int all_years_idxs_tree = all_years_idxs[tree_idxs];
    array[N_years[t]] real years_tree = years[tree_idxs];
    int stand_idx = stand_idxs[t];
    int species_idx = species_idxs[t];
    vector[N_years[t]] gdd_obs_tree = gdd_obs[tree_idxs];
    vector[N_years[t]] sm_obs_tree = sm_obs[tree_idxs];
    vector[N_years[t]] vpd_obs_tree = vpd_obs[tree_idxs];
    vector[N_years[t]] mu =  alpha
    + beta_gdd[species_idx] * (gdd_obs_tree - gdd0)
    + beta_sm[species_idx] * (sm_obs_tree - sm0)
    + beta_vpd[species_idx] * (vpd_obs_tree - vpd0)
    + f_sh[stand_idx, all_years_idxs_tree];
    // mixture weight for shock
    real mw_shock = phi_sck[stand_idx]*omega_conc_sck + (1-phi_sck[stand_idx])*omega_nonconc_sck;
    for(y in 1:N_years[t]){
      // mixture components
      real log_pshock = log(mw_shock) + normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[y] + f[tree_idxs[y]], eta * sqrt(1 + nu / eta^
      real log_pshock_plus_pnonshock = log_mix(mw_shock,
        normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[y] + f[tree_idxs[y]], eta * sqrt(1 + nu / eta^2)),
        normal_lpdf(log_rw_obs[tree_idxs[y]] | mu[y] + f[tree_idxs[y]], eta));
      // probability of shock state
      real lambda_shock = exp(log_pshock - log_pshock_plus_pnonshock);
      sck_state[tree_idxs[y]] = bernoulli_rng(lambda_shock); // or something like categorical_rng(lambda_shock);?

      if(sck_state[tree_idxs[y]] == 1) {
        log_rw_pred[tree_idxs[y]] = normal_rng(mu[y] + f[tree_idxs[y]], eta * sqrt(1 + nu / eta^2));
      }else{
        log_rw_pred[tree_idxs[y]] = normal_rng(mu[y] + f[tree_idxs[y]], eta);
      }

    }
  }
}
```

Test with a small dataset

Let's run this new model on a very small dataset, 21 trees and 1 site.
The good thing here is that I don't need to tune the (non-)centered parametrization anymore.

```
data <- readRDS(file = file.path(wd, 'output', 'model', 'data_15nov2025_az592.rds'))
data$N_stand_trees <- array(data$N_stand_trees, dim = 1)
data$N_stand_years <- array(data$N_stand_years, dim = 1)
data$stand_start_years_idxs <- array(data$stand_start_years_idxs, dim = 1)
data$stand_trees_end_idxs <- array(data$stand_trees_end_idxs, dim = 1)
data$stand_trees_start_idxs <- array(data$stand_trees_start_idxs, dim = 1)

if(FALSE){
  fit <- stan(file=file.path(wd, 'model', 'stan', 'model9_marginalshocks.stan'),
              data=data, seed=5838293, chains = 4,
              warmup=1000, iter=2024, refresh=10)
  saveRDS(fit, file.path(wd, 'model/shocks/output/marginal', 'fit.rds'))
}else{
```

```
  fit <- readRDS(file.path(wd, 'model/shocks/output/marginal', 'fit.rds'))
}
```

Let's look at the diagnostics:

```
diagnostics <- util$extract_hmc_diagnostics(fit)
util$check_all_hmc_diagnostics(diagnostics)
```

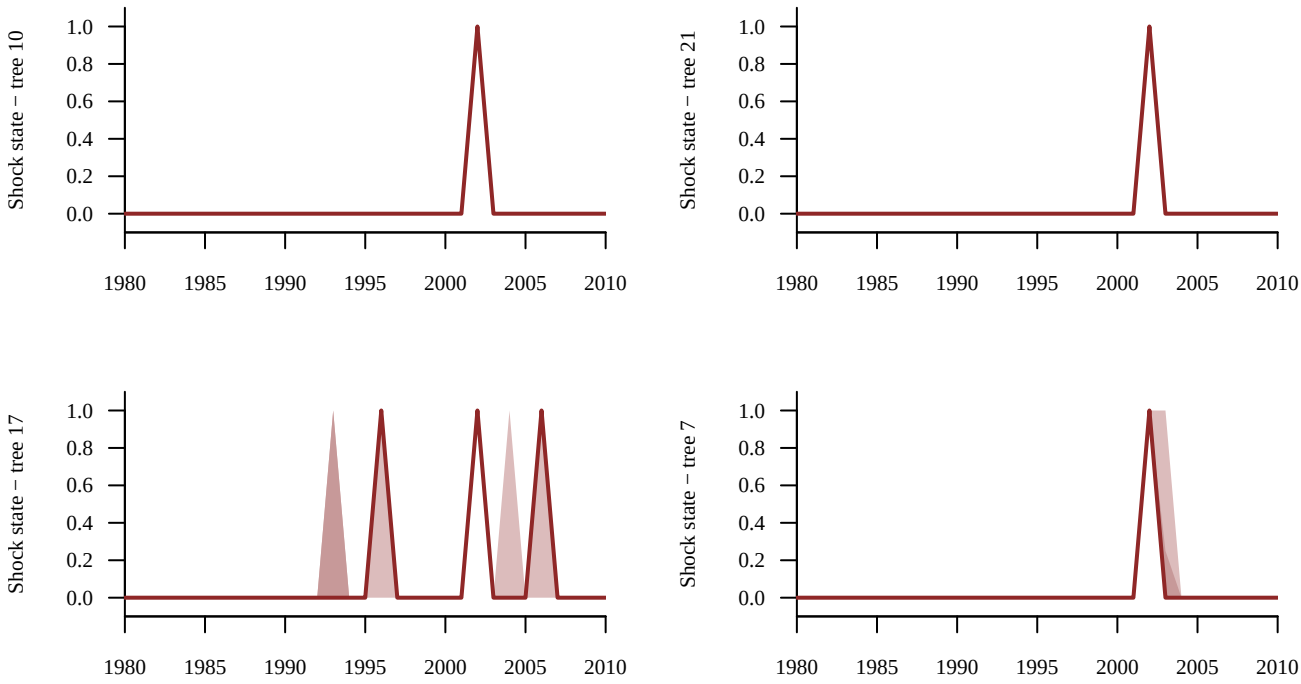
```
  All Hamiltonian Monte Carlo diagnostics are consistent with reliable
Markov chain Monte Carlo.
```

```
samples <- util$extract_expectand_vals(fit)
base_samples <- util$filter_expectands(samples,
                                       c('alpha',
                                         'beta_gdd', 'beta_sm', 'beta_vpd',
                                         'rho_sh', 'gamma_sh',
                                         'rho_sp', 'gamma_sp',
                                         'eta', 'nu',
                                         'phi_sck', 'omega_conc_sck', 'omega_nonconc_sck'),
                                       check_arrays = TRUE)
util$check_all_expectand_diagnostics(base_samples)
```

```
All expectands checked appear to be behaving well enough for reliable
Markov chain Monte Carlo estimation.
```

Clean! Let's continue.
These are the latent shock states:

```
set.seed(123456)
random_trees <- sample(1:data$N_trees, 4)

par(mfrow = c(2,2), mar = c(2, 5, 3, 1))
for(t in random_trees){
  idxs_t <- data$tree_start_idxs[t]:data$tree_end_idxs[t]
  names <- sapply(idxs_t,
                  function(x) paste0('sck_state[', x, ']'))
  util$plot_conn_pushforward_quantiles(samples, names, data$years[idxs_t],
                                       xlab="", ylab=paste0("Shock state - tree ",t),
                                       display_ylim=c(-0.1, 1.1), display_xlim = c(1980, 2010))
}
```

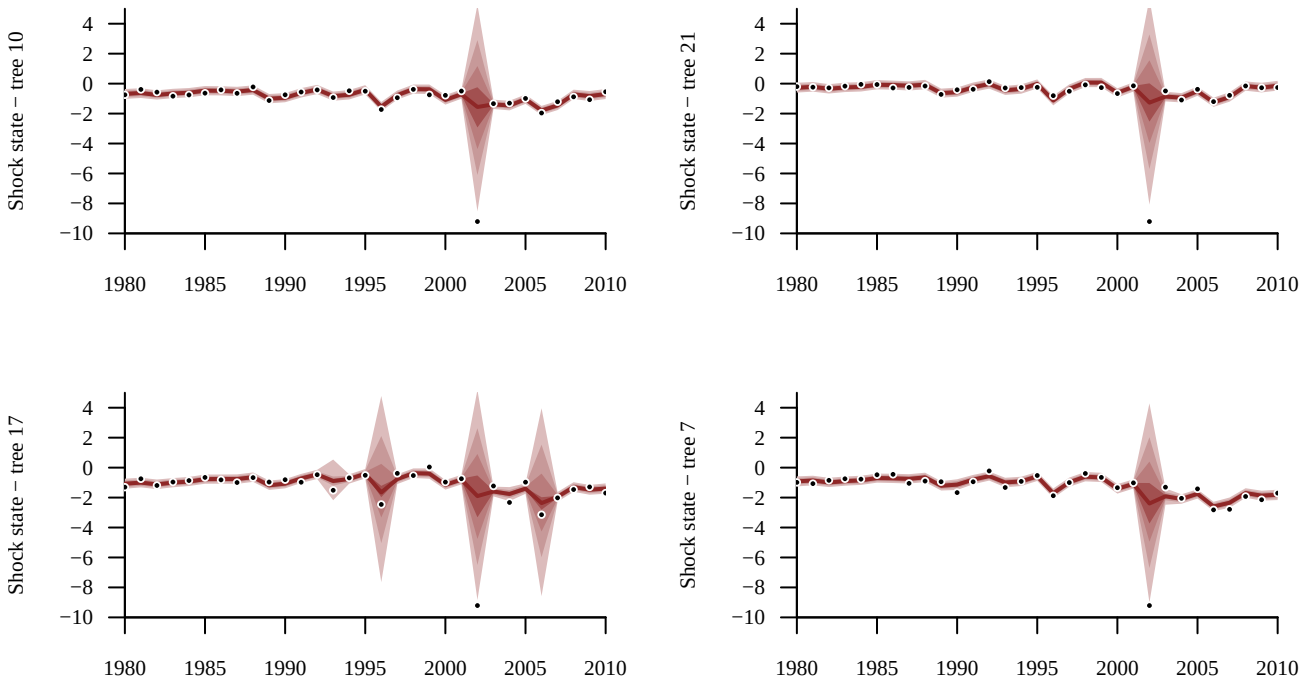


Next step

Is it possible to recover the posterior of $\delta_{\text{shock}}(k) \mid y(k)$, without having $\tau_{\text{inner}}^{\text{shock}}$ or $\tau_{\text{outer}}^{\text{shock}}$?
Maybe a first step could be to fix: $\tau_{\text{inner}}^{\text{shock}} = 0$?

Because for now my predictions look like:

```
par(mfrow = c(2,2), mar = c(2, 5, 3, 1))
for(t in random_trees){
  idxs_t <- data$tree_start_idxs[t]:data$tree_end_idxs[t]
  names <- sapply(idxs_t,
                  function(x) paste0('log_rw_pred[', x, ']'))
  util$plot_conn_pushforward_quantiles(samples, names, data$years[idxs_t],
                                       xlab="", ylab=paste0("Shock state - tree ",t),
                                       display_ylim=c(-10, 5), display_xlim = c(1980, 2010))
  points(data$years[idxs_t], data$log_rw_obs[idxs_t], pch=16, cex=0.7, col="white")
  points(data$years[idxs_t], data$log_rw_obs[idxs_t], pch=16, cex=0.4, col="black")
}
```



Assuming that tau_inner = 0

If I assume that $\tau_{\text{inner}}^{\text{shock}} = 0$, I can reconstruct the large shocks using the normal-normal conjugancy (*merci* Google):

$$\delta_{\text{shock}}(k) \mid y_k \sim \text{normal}\left(({\tau_{\text{outer}}^{\text{shock}}}^2/({\tau_{\text{outer}}^{\text{shock}}}^2 + \sigma^2)) \times y_k \ , \ \sqrt{({\tau_{\text{outer}}^{\text{shock}}}^2 \times \sigma^2)/({\tau_{\text{outer}}^{\text{shock}}}^2 + \sigma^2)} \right)$$

In the generated quantities block, we can then have:

```
writeLines(
  readLines(file.path(wd, 'model', 'stan',
                      'model9_marginalshocks_zerotauinner.stan'))[395:405])
```

```
      if(sck_state[tree_idxs[y]] == 1) {
        // we can reconstruct shock posterior using the normal-normal conjugancy
        real conjugate_mean = (outer_tau_sck^2 / (outer_tau_sck^2 + sigma^2)) * log_rw_obs[tree_idxs[y]];
        real conjugate_sd   = sqrt((outer_tau_sck^2 * sigma^2) / (outer_tau_sck^2 + sigma^2));
        delta_sck[tree_idxs[y]] = normal_rng(conjugate_mean, conjugate_sd);
        log_rw_pred[tree_idxs[y]] = normal_rng(mu[y] + f[tree_idxs[y]] + delta_sck[tree_idxs[y]], sigma);
      }else{
        log_rw_pred[tree_idxs[y]] = normal_rng(mu[y] + f[tree_idxs[y]], sigma);
      }
```

```
if(FALSE){
  fit2 <- stan(file=file.path(wd, 'model', 'stan', 'model9_marginalshocks_zerotauinner.stan'),
               data=data, seed=5838293, chains = 4,
               warmup=1000, iter=2024, refresh=10)
  saveRDS(fit2, file.path(wd, 'model/shocks/output/marginal', 'fit2.rds'))
}else{
  fit2 <- readRDS(file.path(wd, 'model/shocks/output/marginal', 'fit2.rds'))
}
```

Diagnostics are cleaned:

```
diagnostics <- util$extract_hmc_diagnostics(fit2)
util$check_all_hmc_diagnostics(diagnostics)
```

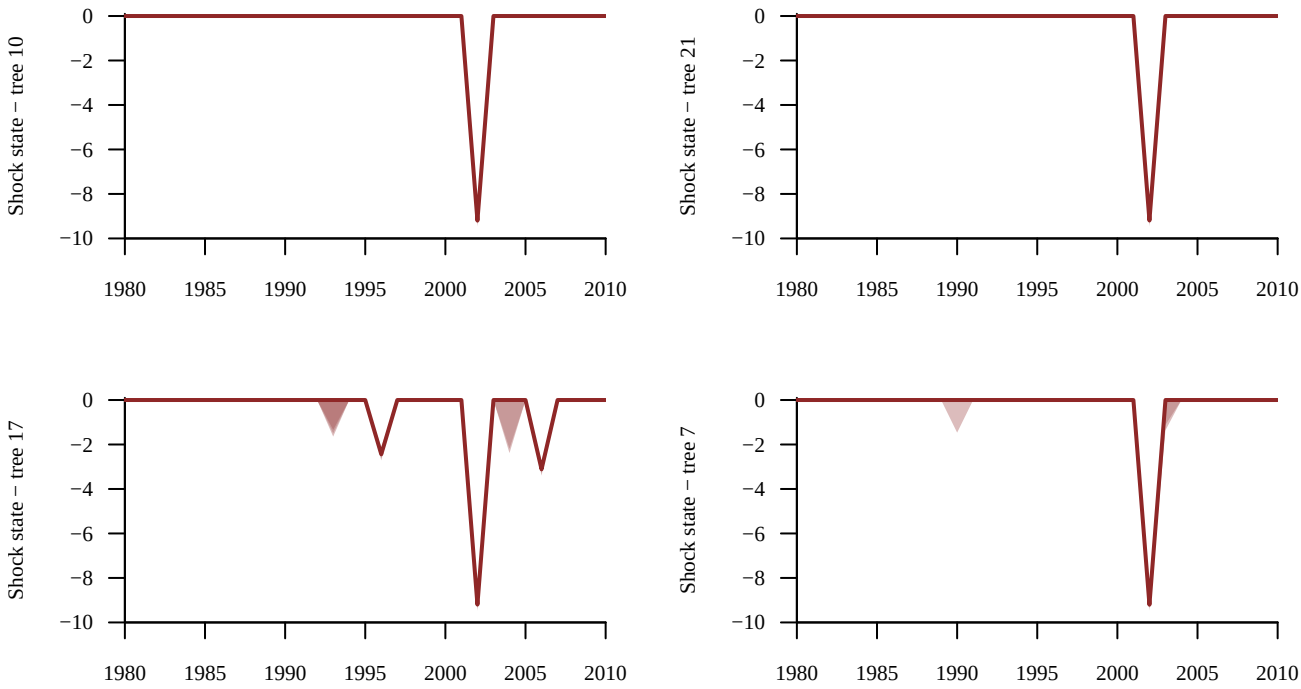
```
  All Hamiltonian Monte Carlo diagnostics are consistent with reliable
Markov chain Monte Carlo.
```

```
samples <- util$extract_expectand_vals(fit2)
base_samples <- util$filter_expectands(samples,
                                       c('alpha',
                                         'beta_gdd', 'beta_sm', 'beta_vpd',
                                         'rho_sh', 'gamma_sh',
                                         'rho_sp', 'gamma_sp',
                                         'sigma', 'outer_tau_sck',
                                         'phi_sck', 'omega_conc_sck', 'omega_nonconc_sck'),
                                       check_arrays = TRUE)
util$check_all_expectand_diagnostics(base_samples)
```

```
All expectands checked appear to be behaving well enough for reliable
Markov chain Monte Carlo estimation.
```

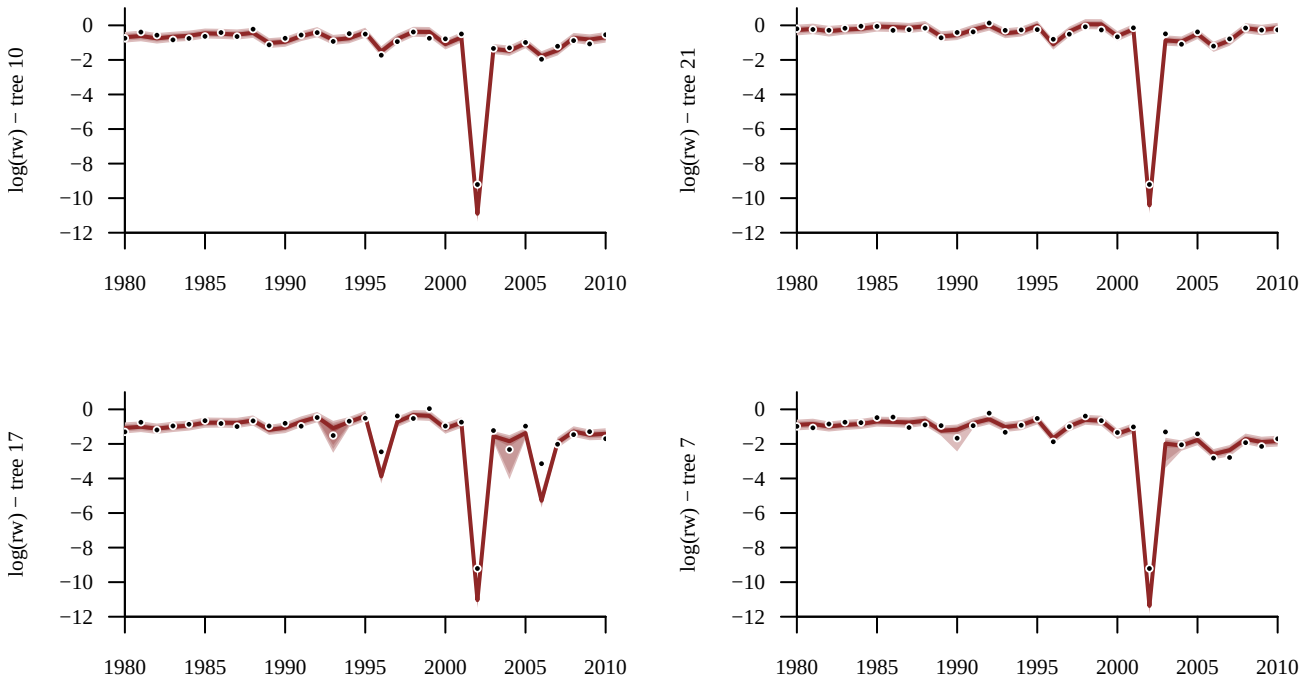
We can look at the latent shocks:

```
par(mfrow = c(2,2), mar = c(2, 5, 3, 1))
for(t in random_trees){
  idxs_t <- data$tree_start_idxs[t]:data$tree_end_idxs[t]
  names <- sapply(idxs_t,
                  function(x) paste0('delta_sck[', x, ']'))
  util$plot_conn_pushforward_quantiles(samples, names, data$years[idxs_t],
                                       xlab="", ylab=paste0("Shock state - tree ",t),
                                       display_ylim=c(-10, 0.1), display_xlim = c(1980, 2010))
}
```



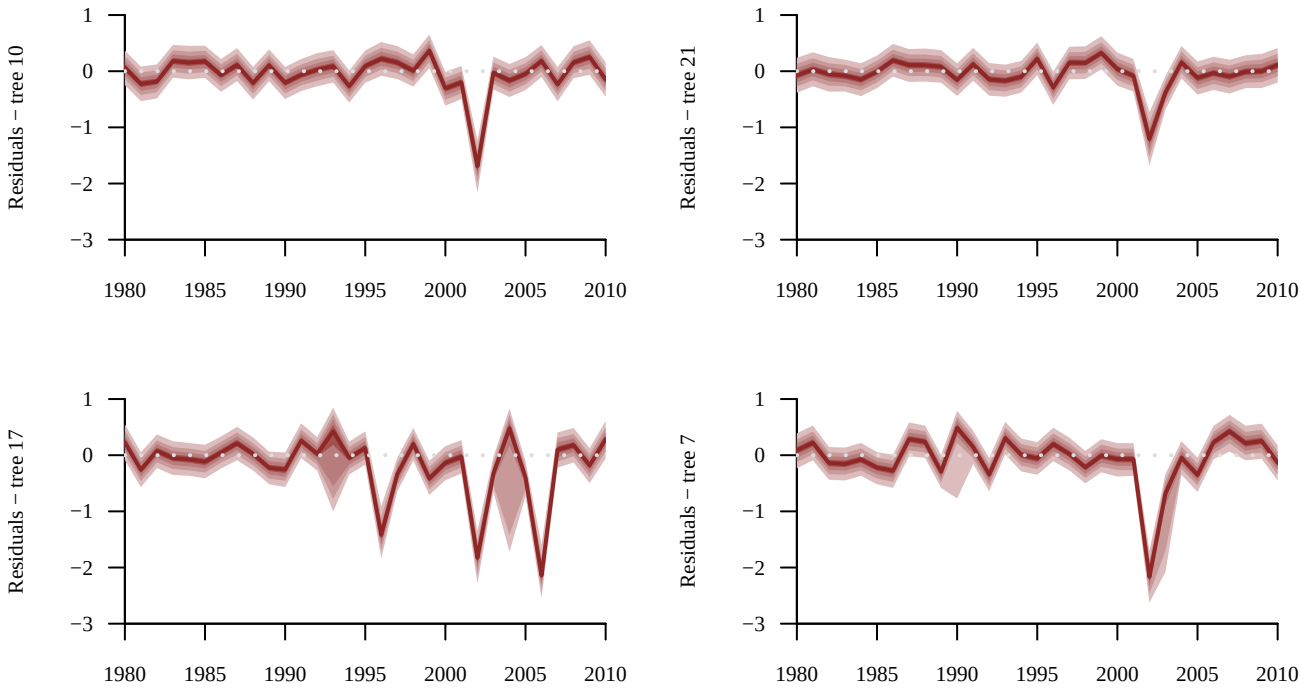
And at the predictions:

```
par(mfrow = c(2,2), mar = c(2, 5, 3, 1))
for(t in random_trees){
  idxs_t <- data$tree_start_idxs[t]:data$tree_end_idxs[t]
  names <- sapply(idxs_t,
                  function(x) paste0('log_rw_pred[', x, ']'))
  util$plot_conn_pushforward_quantiles(samples, names, data$years[idxs_t],
                                       xlab="", ylab=paste0("log(rw) - tree ",t),
                                       display_ylim=c(-12, 1), display_xlim = c(1980, 2010))
   points(data$years[idxs_t], data$log_rw_obs[idxs_t], pch=16, cex=0.7, col="white")
  points(data$years[idxs_t], data$log_rw_obs[idxs_t], pch=16, cex=0.4, col="black")
}
```



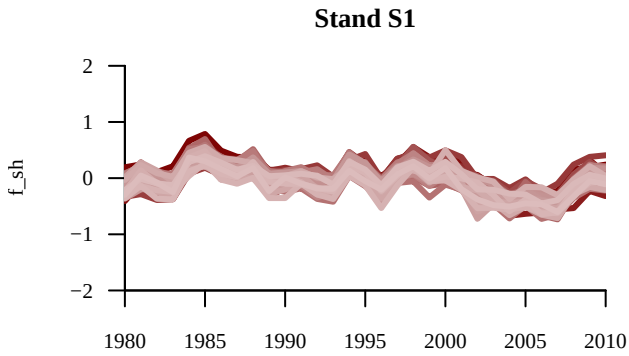
However, it seems that the shocks are bit too large...

```
par(mfrow = c(2,2), mar = c(2, 5, 3, 1))
for(t in random_trees){
  idxs_t <- data$tree_start_idxs[t]:data$tree_end_idxs[t]
  names <- sapply(idxs_t,
                  function(x) paste0('log_rw_pred[', x, ']'))
  util$plot_conn_pushforward_quantiles(samples, names, data$years[idxs_t],
                                       baseline_values = data$log_rw_obs[idxs_t],
                                       xlab="", ylab=paste0("Residuals - tree ",t),
                                       display_ylim=c(-3, 1), display_xlim = c(1980, 2010),
                                       residual = TRUE)
}
```



And, let's not forget the stand-level GP, which is still wiggly:

```
s <- 1
names <- sapply(1:data$N_all_years,
                  function(y) paste0('f_sh[', s, ',', y, ']'))
util$plot_realizations(samples, names, plot_xs = data$all_years[1:data$N_all_years], N_plots = 50,
                       main = paste0('Stand ', data$uniq_stand_ids[s]), ylab = 'f_sh',
                       display_xlim = data$all_years[c(1,data$N_all_years)], display_ylim = c(-2,2))
```



More data!

Let's expand this stuff to moooore data!
914 trees, this take more than 7 hours to run.

```
data <- readRDS(file = file.path(wd, 'output/model', 'data_15nov2025_azPIPO.rds'))

if(FALSE){
  fit3 <- stan(file=file.path(wd, 'model', 'stan', 'model9_marginalshocks_zerotauinner.stan'),
            data=data, seed=5838293, chains = 4,
            warmup=1000, iter=2024, refresh=10)
  saveRDS(fit3, file.path(wd, 'model/shocks/output/marginal', 'fit3.rds'))
}else{
  fit3 <- readRDS(file.path(wd, 'model/shocks/output/marginal', 'fit3.rds'))
}
```

```
diagnostics <- util$extract_hmc_diagnostics(fit3)
util$check_all_hmc_diagnostics(diagnostics)
```

```
  All Hamiltonian Monte Carlo diagnostics are consistent with reliable
Markov chain Monte Carlo.
```

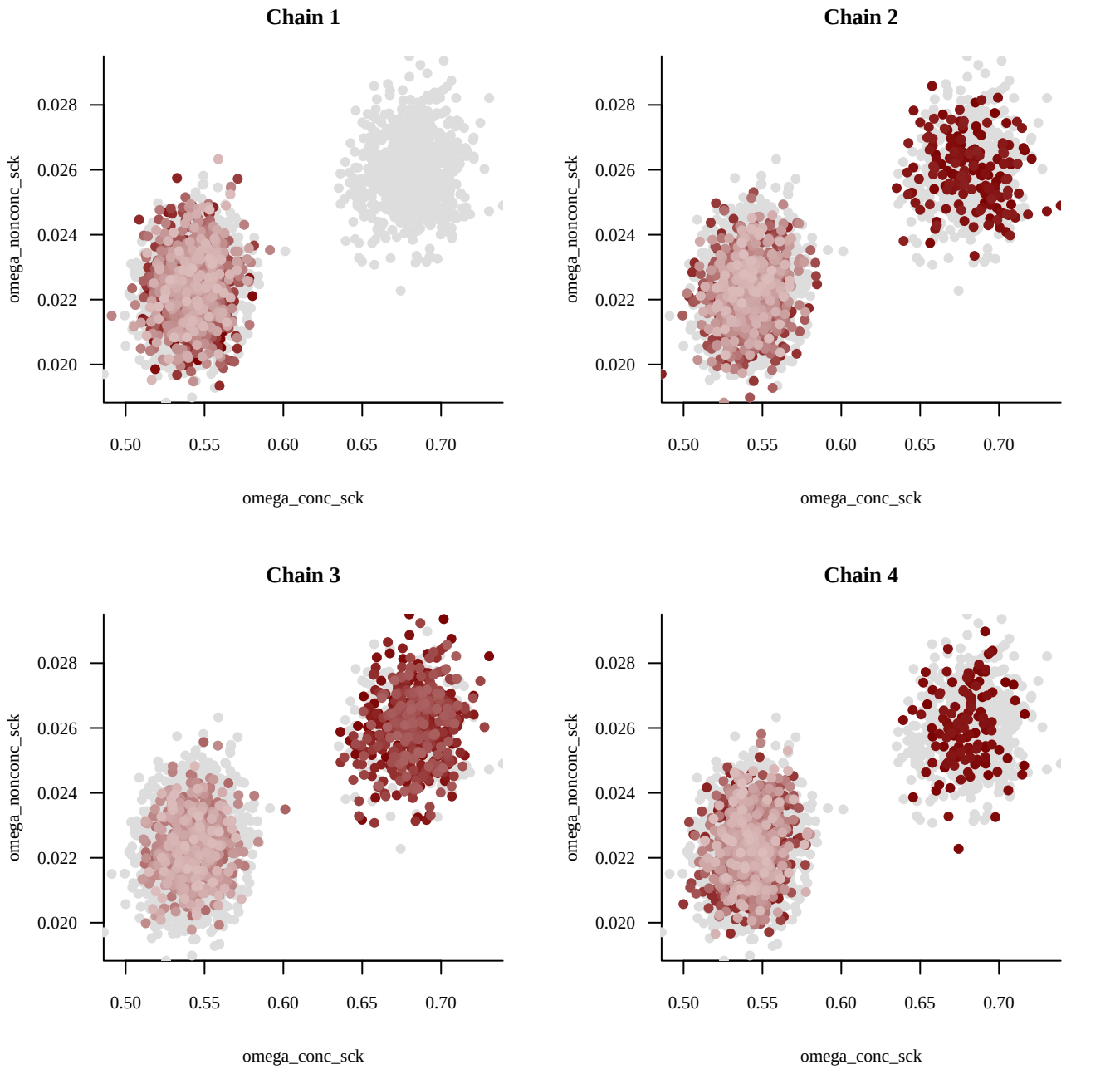
```
samples <- util$extract_expectand_vals(fit3)
base_samples <- util$filter_expectands(samples,
                                       c('alpha',
                                         'beta_gdd', 'beta_sm', 'beta_vpd',
                                         'rho_sh', 'gamma_sh',
                                         'rho_sp', 'gamma_sp',
                                         'sigma', 'outer_tau_sck',
                                         'phi_sck', 'omega_conc_sck', 'omega_nonconc_sck'),
                                       check_arrays = TRUE)
util$check_all_expectand_diagnostics(base_samples)
```

```
alpha:
  Chain 2: hat{ESS} (94.815) is smaller than desired (100).
  Chain 3: hat{ESS} (96.226) is smaller than desired (100).

beta_gdd[1]:
  Chain 2: hat{ESS} (83.445) is smaller than desired (100).

beta_sm[1]:
  Chain 2: hat{ESS} (70.945) is smaller than desired (100).

rho_sp[1]:
  Chain 1: hat{ESS} (76.078) is smaller than desired (100).
  Chain 2: hat{ESS} (37.400) is smaller than desired (100).
  Chain 3: hat{ESS} (56.188) is smaller than desired (100).
  Chain 4: hat{ESS} (65.237) is smaller than desired (100).

sigma:
  Chain 2: hat{ESS} (38.755) is smaller than desired (100).
  Chain 3: hat{ESS} (36.254) is smaller than desired (100).
  Chain 4: hat{ESS} (89.635) is smaller than desired (100).

outer_tau_sck:
  Chain 3: hat{ESS} (46.249) is smaller than desired (100).

phi_sck[12]:
  Chain 3: hat{ESS} (68.030) is smaller than desired (100).

omega_conc_sck:
  Chain 2: Right tail hat{xi} (0.871) exceeds 0.25.
  Chain 4: Right tail hat{xi} (0.657) exceeds 0.25.
  Split hat{R} (1.685) exceeds 1.1.
  Chain 2: hat{ESS} (7.744) is smaller than desired (100).
  Chain 3: hat{ESS} (4.191) is smaller than desired (100).
  Chain 4: hat{ESS} (10.152) is smaller than desired (100).

omega_nonconc_sck:
  Split hat{R} (1.353) exceeds 1.1.
  Chain 2: hat{ESS} (10.463) is smaller than desired (100).
  Chain 3: hat{ESS} (5.389) is smaller than desired (100).
  Chain 4: hat{ESS} (15.782) is smaller than desired (100).

Large tail hat{xi}s suggest that the expectand might not be
sufficiently integrable.

Split Rhat larger than 1.1 suggests that at least one of the Markov
```

```
chains has not reached an equilibrium.

Small empirical effective sample sizes result in imprecise Markov chain
Monte Carlo estimators.
```

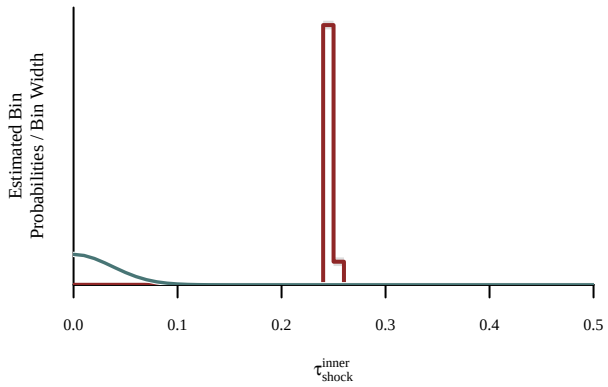
The model clearly wants to concentrate on lower values of $\omega_{\text{shock}}^{\text{concordant}}$.

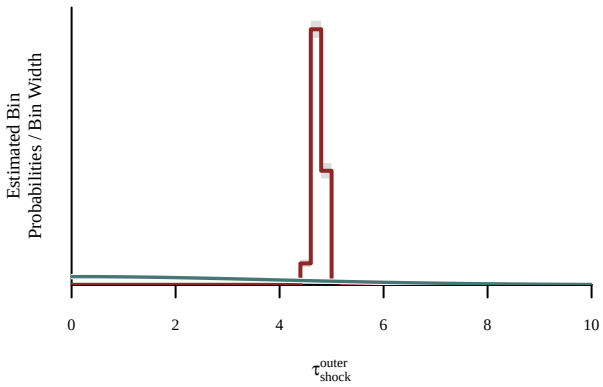
```
util$plot_pairs_by_chain(samples[['omega_conc_sck']], 'omega_conc_sck',
                         samples[['omega_nonconc_sck']], 'omega_nonconc_sck')
```

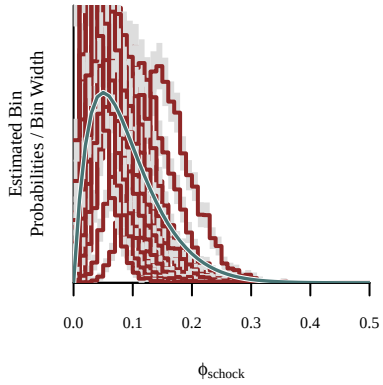


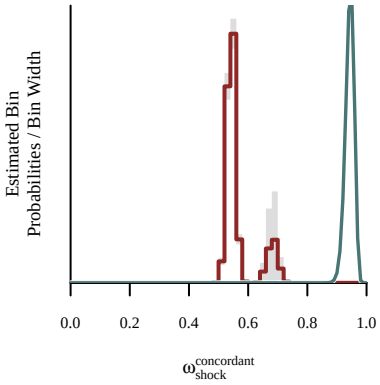
If we look at all shock-related parameters, we can see that both modes for $\omega_{\text{shock}}^{\text{concordant}}$ are very far from our prior domain expertise.

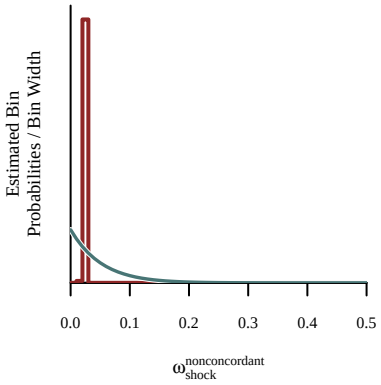
```
# Quickly look at posteriors
nf <- layout(
  matrix(c(1,1,1,2,2,2,
           3,3,4,4,5,5),
         ncol=6, byrow=TRUE)
)
util$plot_expectand_pushforward(samples[['sigma']], 50,
                                flim = c(0,0.5),
                                display_name=expression(tau[shock]^inner))
xs <- seq(0, 2, 0.01)
ys <- dnorm(xs, 0, log(1.1) / 2.57)
lines(xs, ys, lwd=2.7, col='white')
lines(xs, ys, lwd=1.7, col=util$c_mid_teal)

util$plot_expectand_pushforward(samples[['outer_tau_sck']], 50,
                                flim = c(0,10),
                                display_name=expression(tau[shock]^outer))
xs <- seq(0, 10, 0.01)
ys <- dnorm(xs, 0, 10 / 2.57)
lines(xs, ys, lwd=2.7, col='white')
lines(xs, ys, lwd=1.7, col=util$c_mid_teal)

util$plot_expectand_pushforward(samples[['phi_sck[1]']], 50,
                                flim = c(0,0.5),
                                display_name=expression(phi[schock]))
for(s in 2:data$N_stands){
  util$plot_expectand_pushforward(samples[[paste0('phi_sck[',s,']')]], 50,
                                  flim = c(0,0.5),
                                  display_name=expression(phi[schock]), add = T)
}
xs <- seq(0, 0.5, 0.01)
ys <- dbeta(xs, 2, 20)
lines(xs, ys, lwd=2.7, col='white')
lines(xs, ys, lwd=1.7, col=util$c_mid_teal)

util$plot_expectand_pushforward(samples[['omega_conc_sck']], 50,
                                flim = c(0,1),
                                display_name=expression(omega[shock]^concordant))
xs <- seq(0, 1, 0.01)
ys <- dbeta(xs, 230, 14)
lines(xs, ys, lwd=2.5, col='white')
lines(xs, ys, lwd=1.7, col=util$c_mid_teal)
util$plot_expectand_pushforward(samples[['omega_nonconc_sck']], 50,
                                flim = c(0,0.5),
                                display_name=expression(omega[shock]^nonconcordant))
xs <- seq(0, 1, 0.01)
ys <- dbeta(xs, 1, 20)
lines(xs, ys, lwd=2.7, col='white')
lines(xs, ys, lwd=1.7, col=util$c_mid_teal)
```


Estimated Bin Probabilities / Bin Width
0.0
0.1
0.2
0.3
0.4
0.5
τ_{shock}^{inner}


Estimated Bin Probabilities / Bin Width
0
2
4
6
8
10
τ_{shock}^{outer}


Estimated Bin Probabilities / Bin Width
0.0
0.1
0.2
0.3
0.4
0.5
ϕ_{schock}


Estimated Bin Probabilities / Bin Width
0.0
0.2
0.4
0.6
0.8
1.0
$\omega_{shock}^{concordant}$


Estimated Bin Probabilities / Bin Width
0.0
0.1
0.2
0.3
0.4
0.5
$\omega_{shock}^{nonconcordant}$